

Group Design Project

Autonomous Nano-Drone Delivery

Hardware Team - ADD3



Coordinator: Prof. Mirko Kovac
Supervisors: Prof. Ferri Aliabadi, Dr. Elnaz Naghibi
Department: Department of Aeronautics
Course: H401 MEng. Aeronautical Engineering
Module: AERO96005 - Group Design Project
Academic year: 2020/2021

Student: Ryo Haga
CID: 01341504
Date: 14/06/2021

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ, U.K.

Abstract

Using unmanned aerial vehicles (UAV) on environmental monitoring is becoming increasingly demanding and popular over the past years. The aim of this Group Design Project is to develop a team of DJI Tello drones that can navigate through the forest until it identified a target, and then collect soil and leaf samples, as well as dropping and picking up a sensor package. The main objective of the Hardware team was to design modular mechanisms that can be integrated onto the drone and operate one of the preset tasks. Through literature reviews, brainstorming, mathematical modelling and experimental testing of prototypes, four mechanisms, a modular attachment base and other supporting parts were designed and built. The designed mechanisms reached Technology Readiness Level 4 on multiple tasks included in the mission profile: namely soil sampling collection, drop-off and pick-up of the sensor package. Additionally, when mounted on the drone, the mechanisms achieved a success rate of 70% or greater. However, the project did not reach the stage of testing the entire mission profile. Further improvements of the hardware are in sight, and is left for future investigations.

Project Team

Sub-Team	Member	Role
ADD1: Project Management	Yekaterina Goodwin Harrison Moss	Project Manager (CEO) Project Manager (CTO)
ADD2: Machine Learning and Navigation	Jules Baldantoni Dylan Hughes Jake Lewis Natasha Lim Su Jiat Ong Antony Yang	Computer Vision Sub-Team Lead & Integration Machine Learning & Computer Vision Navigation (Exploration) Navigation (Pathfinding) Control
ADD3: Hardware Development	Yinka Adekoya Carlotta Berto Salmeron Ryo Haga Alexandru-Teodor Popescu Ti Sum Yuan Dong Wang Ruan	Sub-Team Lead & Soil Sampling Soil Sampling Soil Sampling & Payload Pick-Up Payload Pick-Up Payload Drop-Off Leaf & Soil Sampling
ADD4: Perception and State Estimation	Daniel Ali Ben-Tiba Anahita Laverack Jiaying Li Logan Sritharan Cameron Stanislawski-Doyle Siu Yuen	Sub-Team Lead & SLAM LIDAR & Mapping Collision Detection Navigation Simulation Collision Avoidance Visual Inertial Odometry
ADD5: Experimental Set-Up, Safety and Rapid Prototyping	Qiyang Deng Quanyi Dong Ziwen Meng Jiayi Sun Wei Tin Hanzhi Zhang	Safety & Manufacturing Safety & Simulation Simulation Lead Simulation Sub-Team Lead & Experiment Design Simulation

My contribution was mainly on the development of the soil sampling mechanism, making a prototype for the drop-off mechanism and the performance analysis of the pick-up mechanism.

Acknowledgements

Firstly, I would like to thank the board of supervisors and Dr. Basaran Kocer for the constructive feedback and the strong support. Discussing miscellaneous ideas on potential solutions of the mechanisms with you gave me great inspirations, and it was very reassuring to have experienced academics to ask for advice.

I would also like to thank my fellow team members in the Hardware team for attending meetings in early morning, making it easier for me to collaborate from GMT +9. Collaborating as a team has been enjoyable, and I am convinced that it was not possible to reach the goal we have met without you.

Finally, I want to thank Antony Yang for being a strong help throughout the project. Collecting flight data during busy times must have been hard, but thank you very much for prioritising my requests and responding swiftly.

Contents

List of Figures	iii
List of Tables	iii
1 Introduction	1
1.1 Motivation	1
1.2 Mission profile	1
1.3 Objectives	1
1.4 Design restrictions	1
2 Common Features Across All Missions	2
2.1 Material selection	2
2.2 Modular attachment	2
3 Soil Sampling Mission	3
3.1 Literature review and design choice	3
3.2 Quantitative Evaluation	3
3.2.1 Analysis on required energy for the cone to penetrate into elasto-plastic media	3
3.2.2 Analysis of acceleration data obtained from the IMU	5
3.3 Dropping tests	5
3.4 Soil sampling claw inspired from leaf sampling	7
4 Sensor Package Mission	8
4.1 Literature review	8
4.2 Prototype of the drop off mechanism	8
4.3 Designing the pick up mechanism	8
4.4 Precision of the pick up mechanism	9
5 Integration and Results	9
5.1 Position of centre of gravity	9
5.2 Summary of mechanisms' performance	10
6 Discussion and Future Improvements	10
6.1 Impact of the pandemic	10
6.2 Further improvement of the design	10
7 Conclusion	12
References	13
A Supplementary data	II
A.1 Magnet precision test	II
B Details of the theory used for soil sampling design	II
C Matlab Code	III
C.1 Function used to analyse landing data from IMU	III
D Candidates of mechanisms and Pugh diagrams	V
E CAD Drawings	VI

List of Figures

1	Overview of the mission profile.	2
2	Geometrical restriction on top of the drone. The edge of the region the rotor goes through during flight is coloured. The unit of the dimensions are in mm	2
3	First prototype of the modular attachment mechanism, inspired from LEGO. Attached and detached with vertical movement.	3
4	Final design of the modular attachment mechanism. Attached and detached with horizontal movement in transverse direction.	3
5	Side view of the cone.	3
6	Comparison between three analytical models	5
7	The time variance of a , v and h of test flights using "Safe landing" command, "Kill" command from 0.4 m and 0.6 m in height.	6
8	First prototype that has been tested, with 30° half-apex angle and six circular holes on the conical surface.	6
9	Five cone mechanisms used for the drop test on dry sand. The half-apex angle α and the shape of the holes of each cone is listed in Table 4.	6
10	Experimental setup of the dropping test of the cone mechanisms. After significant number of trials, ADD03 members became experts of dropping cones.	7
11	Collected mass of sand for various dropping heights, from 10 cm to 25 cm in 5 cm increments. . .	7
12	Claw mechanism designed specifically to collect soil containing organic matter.	8
13	First prototype of the drop off mechanism	8
14	The final design of the drop off mechanism assembled with the payload. On top of the spring, the male part of the modular attachment can be seen.	8
15	Experimental setup. The pick up mechanisms were lowered vertically along the yellow ruler. . .	9
16	The boundaries of the effective region of the pick up mechanism. The boundaries shown on the figure represents the range of the centre of the pick up mechanisms, in local x, y coordinates, that succeeded in picking up the payload.	9
17	Side views and some forward views of assembled CAD of the Tello drone with all sensors and mechanisms integrated.	11
A.1	The boundaries of the effective region and the individual attempts of the pick up mechanism. The boundaries shown on the figure represents the range of the centre of the pick up mechanisms, in local x, y coordinates, that succeeded in picking up the payload.	II
D.2	Pugh table for choosing gripping mechanisms. Magnetic gripper achieved the highest score. . .	V

List of Tables

1	Soil properties, assuming minimal clay and unlubricated guage surface.	4
2	Required penetration energy U_{pen} calculated using three analytical approaches.	5
3	Landing kinetic energy for different landing methods	5
4	The half-apex angle α and the shape of the holes, for five cone mechanisms that were tested. . .	7
5	Summary of how parts were integrated. Shift in centre of gravity after mounting mechanisms and sensors are listed at the bottom of the table. Note that for the drop off and the pickup mechanisms, the payload is included, since the CG shift is larger with payload. "Net moment" refers to clockwise moment viewing from starboard of the drone.	12

1 Introduction

1.1 Motivation

Nowadays, the majority of monitoring and data collection systems for nature are combinations of ground-based measurement, satellite observation and manned airborne sensors[1]. However, over the last decade, the use of unmanned aerial vehicles (UAV) on environmental monitoring is becoming increasingly popular. There are numerous methods designed for variety of needs, such as concentration of pollutants such as oil spills can be evaluated remotely using UV/IR sensing[2], vegetation state of vineyards can be monitored through thermal camera mounted on UAVs[1], physio-chemical data from freshwater environments can be retrieved by collecting water samples using UAVs[3] and wireless sensor networks (WSNs) can be installed in forested environments by shooting out sensors into trees from UAVs[4]. The fact that UAVs are capable of reaching hazardous or inaccessible area by other monitoring vehicles is a notable advantage of environmental monitoring using UAVs. However, it is common that monitoring UAVs are designed specifically to fit one purpose and reducing the take-off weight to below 3kg is still challenging[1].

In this Group Design Project, a team of drones used to identify contaminants in forested environments is designed, using a small, off-shelf drone, DJI Tello¹. The mechanisms and sensors designed to monitor a specific feature of the environment will be either 3D printed or purchased, leading to low-cost solutions for each purpose. The first target environmental contaminants will be ash dieback on ash trees that is becoming a significant issue over the past years[6]. The region where the ash dieback is spread can be monitored taking a small amount of leaf samples and testing in laboratories.² The second target is rather general: issues such as bacterial contamination and soil pollution can be monitored by collecting soil samples[7] or setting up a small monitoring station that contains sensors such as pH sensors. These targets lead to the following mission profile of this project.

1.2 Mission profile

As visually represented in Figure 1³, the mission profile consists of three stages. The first stage is consists of the take-off and the flight towards the sampling location while avoiding obstacles. The second stage is

the sampling process, which is identifying the target and performing a task. The operated task should be one of the following four missions⁴:

1. **Soil sampling:** Collection of at least 1g of top-soil from the ground.
2. **Drop-off:** Drop-off of a sensor package weighing 3g into damp soil.
3. **Pick-up:** Retrieve the sensor package 8 hours after the drop-off.
4. **Leaf sampling:** Collection of at least 0.2g of leaf tissue from living trees.

Once the task has been completed, the drone should return to base, once again, avoiding obstacles.

1.3 Objectives

The task of the ADD03 team was to design modular mechanisms that can be integrated onto DJI Tello and to perform aforementioned tasks. In addition to the designed mechanisms, LIDAR, FPV camera and Arduino board used to navigate the drone ⁵ will also be integrated onto the drone. The sensors were be chosen by the ADD04 team; however, their physical integration was be done by ADD03. Additionally, Computer Aided Design (CAD) and mechanical drawings were made, and is shown in Appendix E.

The preset goal of the ADD03 team was to meet level 4 of the Technology Readiness Levels, which corresponds to ‘component and/or breadboard validation in laboratory experiment’[8].

This report will mainly focus on the soil sampling mission and the sensor package missions.

1.4 Design restrictions

- **Weight and CG** The sum of the additional weight on the Tello should be less than 47g, and the net external moment due to external weight should be below $124.2 \text{ cm} \cdot \text{gf} \approx 1.22 \times 10^{-2} \text{ N m}$.⁶
- **Geometrical restrictions:** Above the body of Tello, the sensors and mechanisms should be located avoiding the rotation of the rotors. Similarly, sensors and mechanisms attached beneath the body should avoid the sight of the height sensor.

⁴Drop-off and pick-up missions will be refereed as "sensor package missions".

⁵Referred as "sensors" in the rest of the report

⁶The numbers were confirmed through physical testing, by other members of ADD03.

¹The specification can be seen here [5].

²Confirmed by ADD01.

³Produced by Alexandru-Teodor Popescu (ADD03).

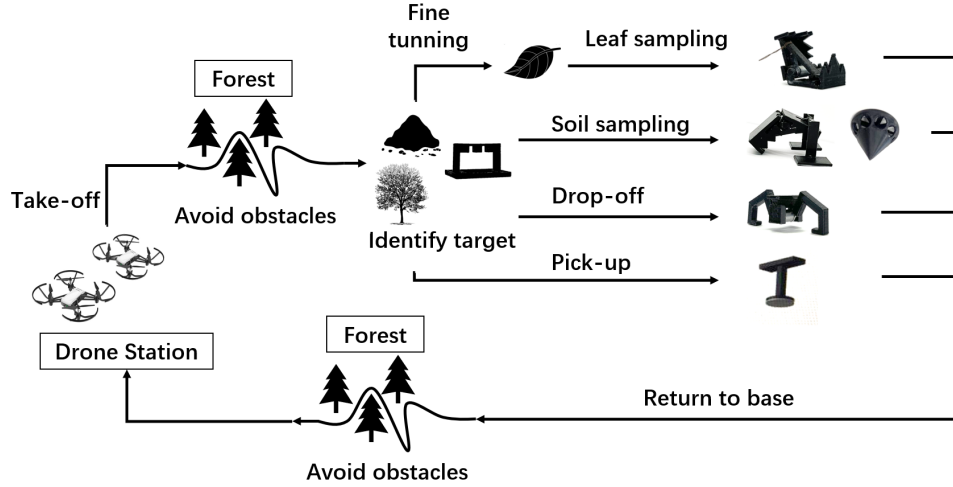


Figure 1: Overview of the mission profile.

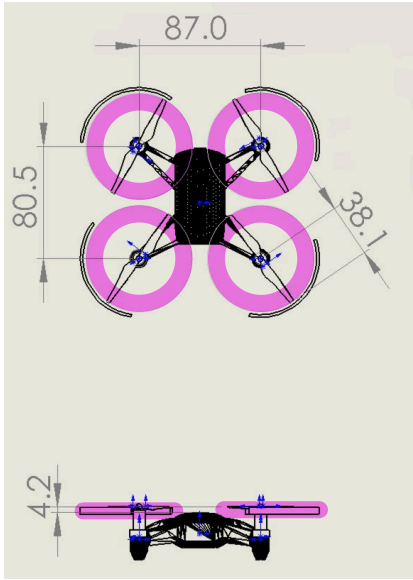


Figure 2: Geometrical restriction on top of the drone. The edge of the region the rotor goes through during flight is coloured. The unit of the dimensions are in mm

- **Manufacturability:** Due to the precision of 3D printing using PLA, the thickness of any parts should be greater than 0.8mm and the hole diameter should be greater than 0.3 mm.⁷
- **Energy supply:** Mechanisms that does not require power supply from the drone is strongly recommended to avoid reducing the flight time.

⁷The values were given from the ADD05 team.

2 Common Features Across All Missions

2.1 Material selection

In this Group Design project, excluding few exceptions such as sensors, springs and magnets, most of the parts were 3D printed to acquire the advantage of rapid prototyping. Hence there were two available options for the material choice: polylactic acid (PLA) and acrylonitrile butadiene (ABS).

The order of masses and impact under consideration in our missions were considerably small compared to the yield strength and other mechanical properties to be evident. Hence, choosing PLA to shorten the printing time was prioritised compared to the enhance impact resistance of ABS, due to the short time period allowed for this project. [△] cite

2.2 Modular attachment

In every mission, sensors are mounted onto the drone since it is essential for navigation. Then, depending on the mission, the corresponding mechanism is attached onto a modular mounting base, which is attached to the drone itself to ensure flexibility of the drone mission. This modular mounting base allows switching of mechanisms and significantly increases flexibility of the drone missions

The modular attachment should hold the mechanism during the flight and mission, but should be easily attachable by hand for convenience. At first, a LEGO-like attachment was considered, as shown in Figure 3, attaching direction is parallel to the direction of largest

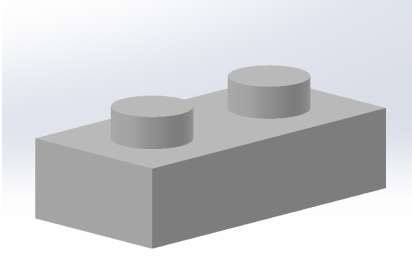


Figure 3: First prototype of the modular attachment mechanism, inspired from LEGO. Attached and detached with vertical movement.

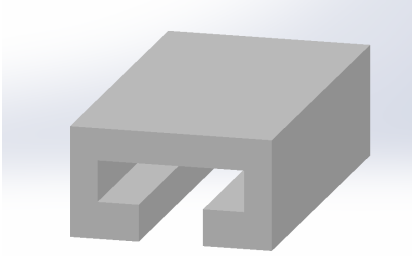


Figure 4: Final design of the modular attachment mechanism. Attached and detached with horizontal movement in transverse direction.

acceleration during flight (vertical direction, from acceleration data collected during flight). Vertical vibration during the flight was considered to be problematic, as well as the strict tolerance requirement. Hence, an improved solution of a slide-in attachment, shown in Figure 4, was chosen to be the final design, since the attaching and detaching motion was in the horizontal direction.

3 Soil Sampling Mission

3.1 Literature review and design choice

The initial step of designing the soil sampling mechanism was to decide on the target type of soil. Soil can be classified as Organic when the weight percentage of the organic matters included in the soil exceeds 20 to 30%, and for non-Organic, or usually called Mineral, soil further classification is done based on the proportion of sand, silt and clay. [9] Due to accessibility and simplicity of the structure, dry sand without organic matter was chosen as the focus of this mission. Investigating the performance of the mechanism on wet, Organic soil was set as a stretch goal at the beginning of the project.

Brainstorming was done to seek for possible solutions

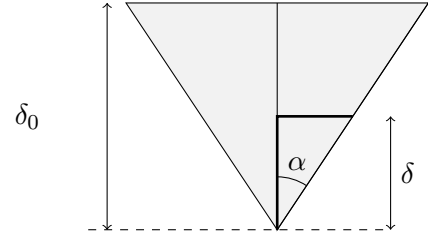


Figure 5: Side view of the cone.

from purely mechanical to bio-inspired solutions, with a strong ambition of applying the Inspire-Abstract-Implement paradigm[10]. However, due to the strict weight restrictions mentioned in Section 1.4, solutions that required actuators such as claws inspired from sea urchin's teeth [11]. Solutions which required specific materials, such as robotic venus flytraps using ionic polymer metal composites [12], were also rejected due to manufacturability.

As a result, a conical mechanism with multiple holes punched through its side was chosen to be designed. Using the kinetic energy of the drone at landing, the cone attached beneath the body was expected to penetrate into the soil, and the plan was to take back the soil that entered through the holes. A conical shape was chosen instead of triangular prism, since the curved surface of the cone reduces the resistance experienced by the cone[13].

3.2 Quantitative Evaluation

3.2.1 Analysis on required energy for the cone to penetrate into elasto-plastic media

To collect enough mass of soil, the cone needs to be submerged into the soil until a certain depth δ_0 , that depends on the geometry of the cone, shown in Figure 5. In this section, the penetration energy U_{pen} is calculated for varying half-apex angle α of the cone using Equation 1 [4], where δ is the penetrated depth of the cone, F_{ind} and F_{fric} represent the indentation force and the friction force, respectively.

$$U_{pen}(\alpha) = \int_0^{\delta_0(\alpha)} F_{ind}(\alpha, \delta) + F_{fric}(\alpha, \delta) d\delta \quad (1)$$

The required penetration depth δ_0 is determined from the volume of the cone that needs to be submerged into the soil, V_{pen} . The target mass of soil is $m_{soil} = 1$ g, hence, using the density value ρ_{sand} from Table 1 the required volume of the soil can be calculated. Note that in this section, loose sand is under consideration. However, considering the position

Table 1: Soil properties, assuming minimal clay and unlubricated guage surface.

Variables	Values
Density of sand, ρ_{sand}	1442 kg m ⁻³ [14]
Yield stress, σ_Y	15 kPa
Strength parameter ratio, $\frac{k_s}{k}$	1×10^{-4}
Soil strength, s_u	1 kPa [15, p.11]
Shear modulus, G	25 MPa [15, p.11]
Gauge roughness, m	$0.7 \in [0, 1]$ [15]
Reference radius, r_b	4.45 mm [16]

of the holes in the conical surface and the thickness of the cone, a safety factor of 4 was multiplied to the required volume of sand, giving

$$V_{pen} = \frac{4m_{soil}}{\rho_{sand}} = \frac{1}{3}\pi\delta_0^3 \tan^2 \alpha.$$

Hence, the required depth δ can be expressed as a function of the half-apex angle α , as shown in Equation 2.

$$\delta_0(\alpha) = \left(\frac{3V_{pen}}{\pi \tan^2 \alpha} \right)^{\frac{1}{3}} \quad (2)$$

To calculate U_{pen} , the forces F_{ind} and F_{fric} need to be expressed as functions of α and δ . Three analytical approaches with slightly different assumptions were compared.

The first approach, which was proposed by Farinha et al.[4], assumes the friction to be half the maximum Coulomb friction, $\sigma_Y/2$, which leads to the uniform indenting pressure of $3\sigma_Y$. Hence, the following Equations 3 and 4 [4] can be substituted into Equation 1 to obtain $U_{pen}(\alpha)$.

$$F_{ind} = 3\sigma_Y \pi (\delta \tan \alpha)^2 \quad (3)$$

$$F_{fric} = \frac{\sigma_Y}{2} 2\pi (\delta_0 \tan \alpha) \delta \quad (4)$$

The original setup of this was a model to analyse cone penetration into trees. In comparison to the original setup, the value of σ_Y is far less reliable since the value greatly depends on the soil that is chosen. The value chosen on Table 1 was chosen from the upper limit to take a conservative approach. Hence, further mathematical analysis was needed.

The second approach is applying the axisymmetric slip-line theory, developed by Xiong et al.[13]. The first approach was the simplification of this axisymmetric slip-line theory, by assuming indentation pressure and

frictional force being independent of α . The variation of U_{pen} with α is expected to be a better representative of reality than the first approach, if the physical parameters were obtained with great accuracy. However, the strength parameter ratio k_s/k , which is equivalent to the ratio of the shear stress at the conical surface to the maximum shear stress in the elasto-plastic media in consideration, was not available for our problem set-up: cone made of PLA penetrating loose sand. Since lower values of k_s/k lead to higher indentation pressure, a conservative value of 1×10^{-4} was taken. Hence, the normal stress $\sigma_n(\alpha, \delta)$ and the shear stress $\tau_n(\alpha, \delta)$ along the generatrix line was calculated using the axisymmetric slip-line theory.

Using Equation 5, the average indentation pressure $\bar{p}$ can be calculated[13]. Note that the upper limit of the integral $\delta' \tan \alpha$ is a non-zero value that is close to 0, set to avoid singularity. According to Xiong, $\bar{p}$ is at minimum for α between 20° and 30° .

$$\bar{p}(\alpha, \delta) = \frac{1}{\pi (\delta \tan \alpha)^2} \int_{\delta_0 \tan \alpha}^{\delta' \tan \alpha} 2\pi [\sigma_n + \tau_n \cot \alpha] d(\delta \tan \alpha) \quad (5)$$

Similarly, average frictional force $\bar{f}$ was calculated using the following equation:

$$\bar{f}(\alpha, \delta) = \frac{1}{2\pi \delta \tan \alpha} \int_{\delta_0 \tan \alpha}^{\delta' \tan \alpha} 2\pi [\sigma_n \cot \alpha + \tau_n] d(\delta \tan \alpha)$$

Hence, the following Equations 6 and 7 can be substituted into Equation 1 to obtain $U_{pen}(\alpha)$.

$$F_{ind} = \bar{p} \pi (\delta \tan \alpha)^2 \quad (6)$$

$$F_{fric} = \bar{f} 2\pi (\delta_0 \tan \alpha) \delta \quad (7)$$

The third approach is to use an analytical model that simulates the penetration of earthworms and expanding plant roots as cones[15]. Although this approach neglects friction, the analysis was based on detailed investigation of soil properties. The soil properties that is used for our purpose is listed in Table 1. The details of the calculation process is included in Appendix B.

For each approach, numerical integration was performed for half-apex angles in the range $\alpha \in [15^\circ, 45^\circ]$, using the matlab inbuilt function `trapz`. The lower bound of α was determined from the restriction due to manufacturability, while the upper bound was determined from the allowable space beneath the body of the drone, so the height sensor does not get covered.

The relationship between U_{pen} and α is shown in Figure 6. Recalling that $\bar{p}$ is at minimum between

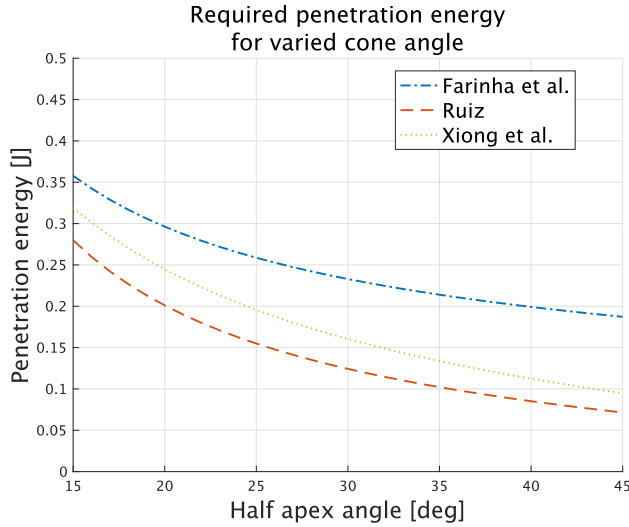


Figure 6: Comparison between three analytical models

Table 2: Required penetration energy U_{pen} calculated using three analytical approaches.

U_{pen} , J	Half-apex angle	
	20°	30°
Farinha	0.300	0.233
Xiong	0.245	0.160
Ruiz	0.201	0.124

$\alpha \in [20^\circ, 30^\circ]$, cones with half-apex angles of 20° and 30° were chosen to be manufactured. The required penetration energy U_{pen} for $\alpha = 20^\circ, 30^\circ$ is listed in Table 2.

3.2.2 Analysis of acceleration data obtained from the IMU

In order to justify whether the kinetic energy at landing is sufficient to overcome U_{pen} , acceleration data obtained from the inertial measuring unit (IMU) on the Tello was recorded during a simple flight following the sequence of take-off, hover and landing commands⁸. The raw data is listed in Appendix A. The recording of linear acceleration in the z -direction a_{raw} , defined as downward positive, was recorded in standard gravitational acceleration unit $g = 9.80665 \text{ m s}^{-2}$. a_{raw} had an offset of acceleration, a_{offset} , which was close to $-1g$ but slightly differed each flight. By taking the average of the raw acceleration before take-off, a_{offset} was obtained. Hence, the upward acceleration a can be obtained from $a = -(a_{raw} - a_{offset})g$ and was in-

⁸IMU data was collected by Antony Yang (ADD02).

Table 3: Landing kinetic energy for different landing methods

Landing methods	Landing kinetic energy, J
"Safe landing"	8.4×10^{-3}
"Kill" from 0.4 m	0.29
"Kill" from 0.6 m	0.43

tegrated once and twice with static initial condition to obtain the velocity v and the altitude h using the matlab inbuilt function `cumtrapz`. The corresponding code is included in Appendix C.1. The time variance of a , v and h are plotted in Figure 7.

It turned out that when the pre-programmed landing command ("Safe landing") is used, the drone decelerates significantly right before landing with an upward acceleration of $a = 1.15 \text{ m s}^{-2}$). This deceleration lead to a significant drop in the kinetic energy, and at landing the kinetic energy was $8.4 \times 10^{-3} \text{ J}$ assuming the overall drone weight of 0.105 kg , a value less than 7% of the nimum prediction of U_{pen} in Table 2.

Therefore, a command that shuts down the motor was developed by ADD02 to increase the landing kinetic energy, which was named "Kill". Once the "Kill" command is executed, either manually or at a preset altitude, the rotation of the rotor decelerates significantly and the drone drops soon after. From the data obtained from the IMU, it turned out that the acceleration a time step after the execution of the "Kill" command was $a = -8.64 \text{ m s}^{-2} \approx 0.88g$ regardless of the altitude of the drone. The time variance of a , v and h for flights with the "Kill" command executed at heights 0.4m and 0.4m are plotted against the "Safe landing" flight in Figure 7. The calculated landing kinetic energy for the "Safe landing" case and with the "Kill" command from different heights are listed in Table 3. It seems that the landing kinetic energy for "Kill" command from 0.4m is sufficient to penetrate the cone mechanism for a required depth, since $U_{pen} < 0.29 \text{ J}$ for all cases listed in Table 2 excluding Farinha for $\alpha = 20^\circ$. In order to confirm the performance of the cones, dropping tests were performed based on the analytical results.

3.3 Dropping tests

The first prototype that has been printed was designed to have $\alpha = 30^\circ$ and six circular holes on the conical surfaces, as shown in Figure 8. A quick dropping test was done on couscous and it was confirmed

Flight profiles with different landing methods

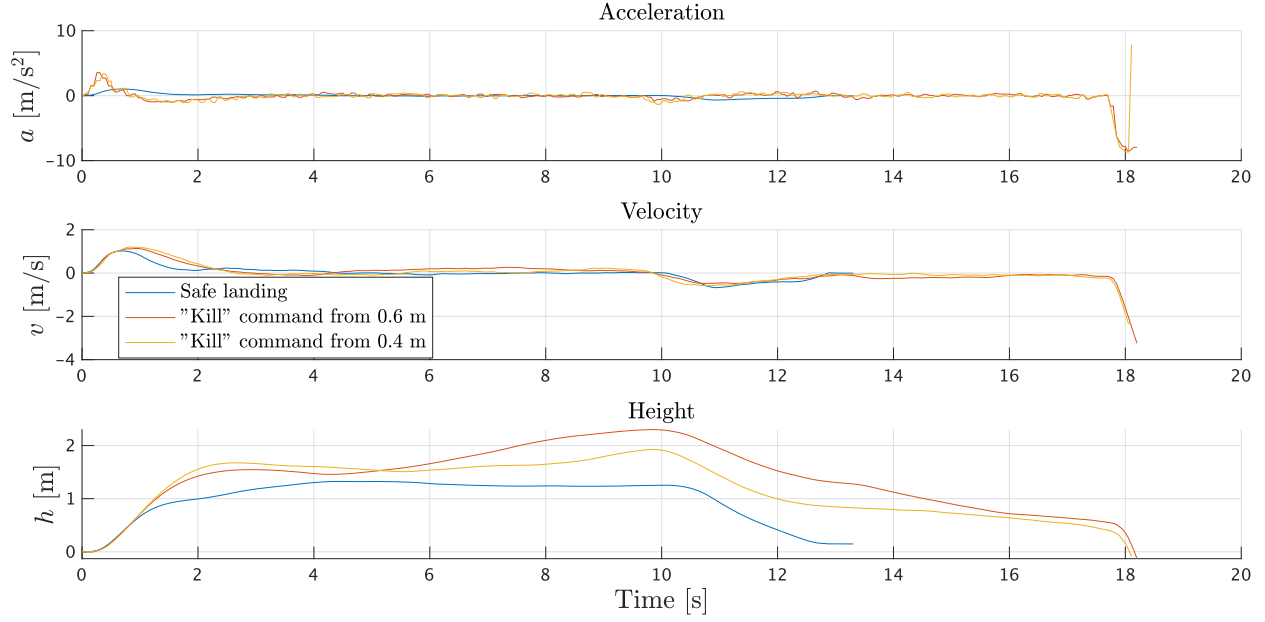


Figure 7: The time variance of a , v and h of test flights using "Safe landing" command, "Kill" command from 0.4 m and 0.6 m in height.

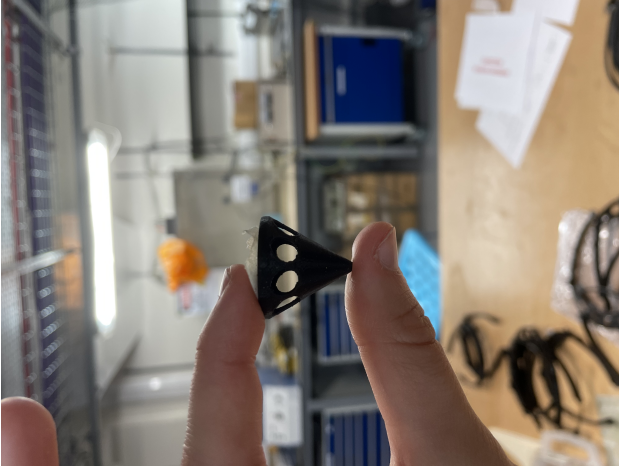


Figure 8: First prototype that has been tested, with 30° half-apex angle and six circular holes on the conical surface.

that the cone mechanism indeed works for dry, granular material with large grain size compared to a typical dry sand. Therefore, the Hardware group decided to design variations of cone mechanisms and choose a design that has the best performance when dropped to dry sand.

In the second design iteration, five cones with various shapes of holes were designed and printed, as shown in Figure 9. The cones were numbered from 1 to 5, and



Figure 9: Five cone mechanisms used for the drop test on dry sand. The half-apex angle α and the shape of the holes of each cone is listed in Table 4.

Table 4 shows the half-apex angle α and the shape of the holes for each cone.

The drop test was conducted based on the following processes:

1. Attach a block weight of approximately 80 g on to the cone. Measure the combined weight using a scale.
2. Drop the integrated cone from 5 cm above dry sand. Gently pick the integrated cone and measure the overall weight. Eject the collected sand from the cone and repeat the dropping for 10 times. The experimental setup is shown in Figure 10.
3. Repeat for dropping heights of 10 cm to 25 cm with

Table 4: The half-apex angle α and the shape of the holes, for five cone mechanisms that were tested.

Parameters	Manufactured cones				
	Cone 1	Cone 2	Cone 3	Cone 4	Cone 5
$\alpha, ^\circ$	30	30	20	20	20
Shape of holes	Circular	Parabolic	Parabolic	Rectangle + semi-circle	Elliptic

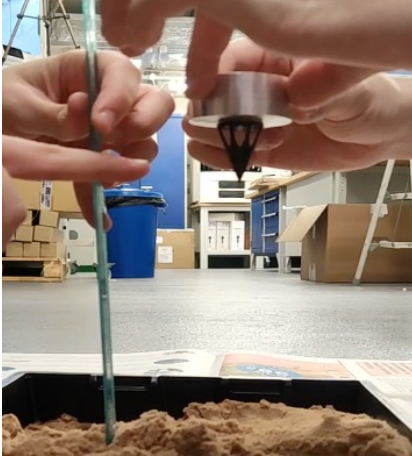


Figure 10: Experimental setup of the dropping test of the cone mechanisms. After significant number of trials, ADD03 members became experts of dropping cones.

5 cm increments.

The variation in the collected sand mass for different dropping heights are shown in Figure 11. The collected mass of sand did not increase as the dropping height increased from 20 cm to 25 cm for most cones, suggesting that the potential energy of the integrated cone at height of 20 cm, 0.206 J, is sufficient to penetrate cone until the lower end of the holes. For all the cones, the collected mass of cones exceeded the target value of 1 g for the dropping height of 20 cm.

It can be seen that Cone 4, with $\alpha = 30^\circ$ and holes in a shape of a rectangle with semi-circles on one end, collected the largest mass of sand for heights higher than 10 cm. Hence, Cone 4 was chosen to be the soil sampling mechanism that is used in environments with loose sand.

3.4 Soil sampling claw inspired from leaf sampling

Additional dropping test was conducted for Cone 4 on wet soil containing some organic matter. It turned

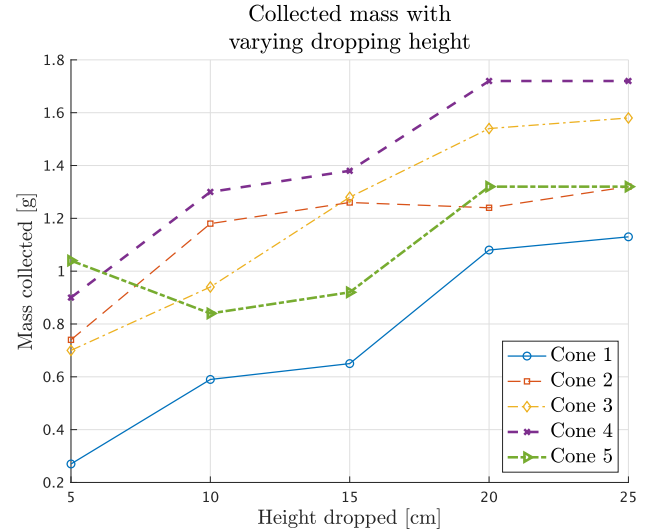


Figure 11: Collected mass of sand for various dropping heights, from 10 cm to 25 cm in 5 cm increments.

out that the cone mechanism does penetrate until a certain depth, however, the mass of soil collected was far less than the target, 1 g. The reason why the cone mechanism did not perform well on wet soil was because less amount of soil tried to restore to its original position once it was displaced by the cone, compared to dry sand that acted more like a fluid. In addition, in some cases big lumps of organic matter clogged the hole of the cone, preventing the surrounding soil to flow into the cone. Hence, an alternative design was required to succeed the soil sampling mission on wet soil.

The claw mechanism that was used in the leaf sampling mission was modified to fit the purpose of the soil sampling mission. The basic idea of the claw mechanism was to prohibit the closing motion of the torsional spring attached to the root of the claw using a stopper, which gets released by a specific external stimulus. In the soil sampling mission, the stopper is let go using the landing impact of the drone. Modifications were made to the direction the stopper gets released, how the claw is mounted on the drone, and landing pads

were added to the stopper to fit to the mission. The printed claw is shown in Figure 12.



Figure 12: Claw mechanism designed specifically to collect soil containing organic matter.

The claw mechanism was tested by pushing the mechanisms into organic soil by hand. As a result, an average of 1.7 g was collected over five trials, which exceeded the target mass of 1 g.

4 Sensor Package Mission

4.1 Literature review

The sensor package missions consists of two distinct tasks: drop-off and pick-up. During the initial stage, variety of gripping mechanisms were considered that can perform both tasks. Pugh analysis was performed and using magnets turned out to be the best solution. The breakdown of the scores in the Pugh analysis can be seen in the Appendix, Table D.2. To avoid using power supply from the drone to deactivate electromagnets, decision was made to design two distinct mechanisms, each for drop-off and pick-up. Hence, permanent magnets were chosen for the pick-up mechanism, and a solution for the drop-off mechanism needed further investigation.

4.2 Prototype of the drop off mechanism

In nature, gripping mechanisms can be classified into friction, adhesion, mechanical interlocking and their combinations[17]. Among these options, mechanical interlocking was able to combined with a pre-loaded torsional spring in order to achieve passivity.

The first prototype of the drop-off mechanism is shown in Figure13. The torsional spring can be attached beneath the drone in a compressed state. A horizontal bar can be attached to the payload to suppress the opening motion, which can also move upwards and release the spring by the reaction force from the ground during landing.

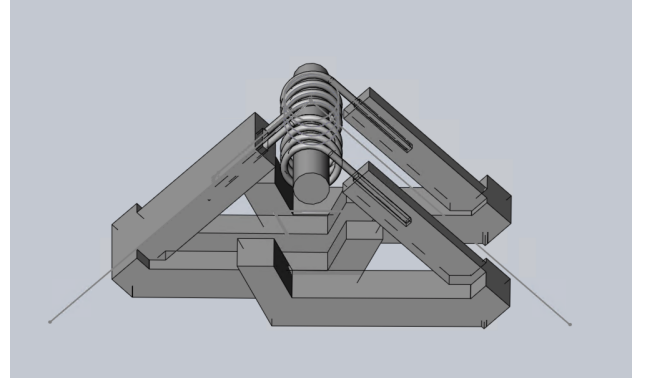


Figure 13: First prototype of the drop off mechanism



Figure 14: The final design of the drop off mechanism assembled with the payload. On top of the spring, the male part of the modular attachment can be seen.

The mechanism was further developed by adjusting the dimensions and designing a suitable payload⁹, and the final design of the mechanism is shown in Figure 14. In parallel to designing the mechanism and the payload, leg extension part was also developed to adjust the height of the drone during landing.

4.3 Designing the pick up mechanism

With the aid of theoretical and experimental performance analysis of small, cheap, off-shelf magnets, F308-20 magnet was chosen to be used for the pick up mechanism among Neodymium magnets of grade 42¹⁰.

Using F308-20 magnets, two pickup mechanisms were designed: one with a single magnet and another with quadruple magnets oriented in a shape of a square.

⁹by Ti Sum (ADD03)

¹⁰by Alexandru-Teodor Popescu (ADD03)



Figure 15: Experimental setup. The pick up mechanisms were lowered vertically along the yellow ruler.

4.4 Precision of the pick up mechanism

During the pick up mission, the drone needs to be navigated towards the payload and land close to the payload. However, analytically predicting what distance the drone can land away from the payload and succeed in sticking the magnets and the steel plate on the payload was a huge challenge. Instead, an experiment was conducted to figure out the required precision of the navigation system of the drone.

The experimental setup is shown in Figure 15. Grid papers of increments of 5 mm and a long ruler was used to guide the vertically movement of the pick up mechanism at a given location. The short side of the steel plate on the payload was defined as the x -direction and the longer side was defined as the y -direction of the local payload coordinate. The centroid of the rectangular plate was defined as the origin of the local coordinate.

The height of the lower surface of the magnet at which the payload attached was recorded 3 times each location. However, the accuracy of the readings were poor due to the instantaneous motion of the payload. Therefore, Boolean data of whether the payload attached to the magnet was used in the data processing. The pick up test was repeated for 32 locations in the first quadrant for singular magnet and 70 locations for quadruple magnets.

From the obtained Boolean data, positive parameters x_0 , y_0 , R_1 and R_2 were determined so that the effective region of the pick up mechanisms can be ex-

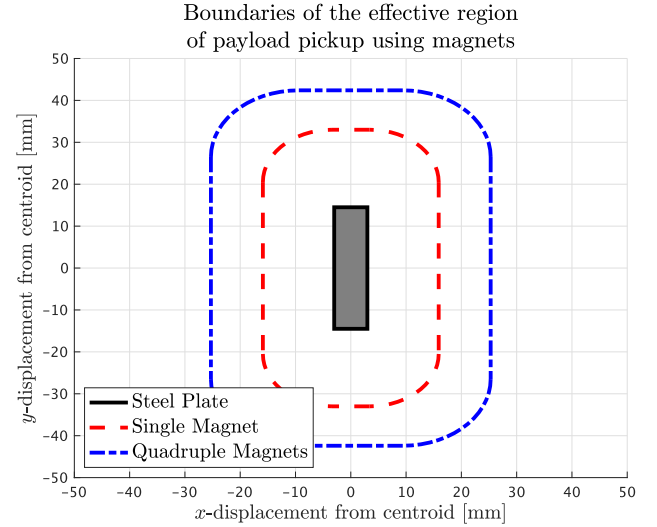


Figure 16: The boundaries of the effective region of the pick up mechanism. The boundaries shown on the figure represents the range of the centre of the pick up mechanisms, in local x, y coordinates, that succeeded in picking up the payload.

pressed using Inequality 8.

$$\max_{(x,y) \in \mathbb{R}_{\geq 0}^2} [\{\max(x - x_0, 0)\}^2 + \{\max(y - y_0, 0)\}^2] \leq R_i^2 \quad (8)$$

Here x_0 and y_0 are common among single and quadruple magnets, where R_1 and R_2 are specific values determined for singular and quadruple magnets, respectively. Inequality 8 comes from the minimum distance of a given point (x, y) to the closest side of a rectangle with centroid at the origin and a vertex at (x_0, y_0) . Note that the right hand side of the inequality is 0 inside the rectangle. Using the determined set of parameters (x_0, y_0, R_1, R_2) , the boundaries of the effective region of the two pick up mechanisms are shown in Figure 16. In the figure, the regions were expanded from $\mathbb{R}_{\geq 0}^2$ to $\mathbb{R}^2$ by taking a mirror reflection along x and y axes. A copy of Figure 16 with the tested coordinates plotted on the same figure is shown in Appendix A.1. The experiment result suggests that by increasing the number of magnets from one to four, the required precision of the navigation for pick up increased by $2(R_2 - R_1) = 19.8$ mm in both x and y direction.

5 Integration and Results

5.1 Position of centre of gravity

Once all the mechanisms has been manufactured and the choice of additional sensors required for navigation

was finalised by ADD04 team, integration was necessary to check the position of the centre of gravity (CG) and the performance of the drone in the integrated state.

With the aid of integrated CADs, which are shown in Figure 17, the total added weight and the shift in CG was calculated, as listed in Table 5¹¹. The method of integration is also summarised on this table.

For all four missions, the total added weights were below 47 g, and the net moment values produced by added components were at most 18% of the limit 124.2 cm·gf. Hence, the weight and CG restrictions were not violated.

5.2 Summary of mechanisms' performance

Through multiple tests, it has been confirmed that the mechanisms were able to perform the tasks on its own. Therefore, one of the pre-set objective, TRL level 4, has been met. Hence, the development of the mechanism moved on to the next step: being tested in the assembled state. The summary of the test results are as follows:

- The cone mechanism collected enough mass of sand for 7 times out of 10 trials. In failed cases, landing orientation was tilted and the loading was no longer axisymmetric.
- The claw mechanism collected enough mass of organic soil for 8 times out of 10 trials. In the failed cases, the landing impact was too large and the stopper disassembled.
- Drop-off mechanism worked 7 times out of 10 trials. In failed cases, the payload flipped over.
- Pick-up mechanism worked 9 times out of 10 trials, under manual control. In the failed case, the landing position was too far away from the payload.

The success rate of the tasks were 70% or above.

6 Discussion and Future Improvements

6.1 Impact of the pandemic

Due to the COVID-19 pandemic situation, the ADD03 team members whom were able to go on campus had very limited access to laboratory instruments.

¹¹The original table produced by Yuan Dong Wang Ruan (ADD03) and then was modified. The used values were investigated by the entire sub-team.

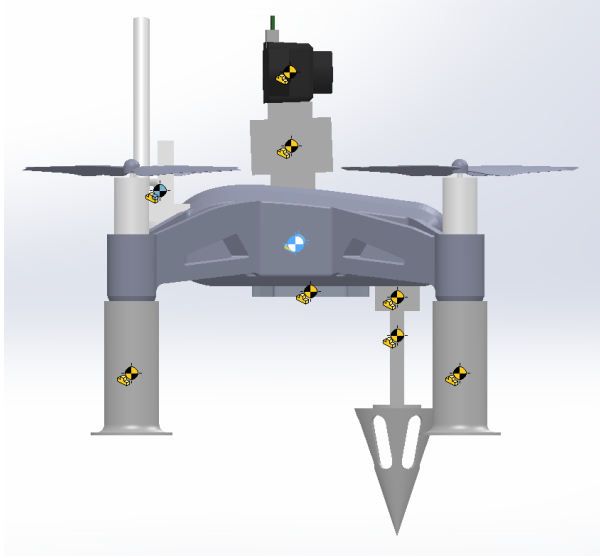
As a member working remotely from abroad, I have proposed multiple experiments, but they were not possible to conduct since the ADD03 team did not have access to pulleys, stands and weights. Here are some experimental validations that could have been taken:

- The dropping test done for the cone mechanism was done by hand. However, the initial plan was to attach the mechanism and the weight to a string, and lift it up using a pulley. The noticeable advantage of this method is that the reproducibility of the dropping procedure increases significantly. In addition, the accuracy of the dropping height will improve.
- Attach the drop-off mechanism with the payload onto a vertically hung spring with a known spring constant, attached to a stand on the other side. By pulling the payload down for a measured distance and releasing it, the maximum vertical acceleration the mechanism can handle before releasing the payload in midair can be determined. By comparing the obtained acceleration with the IMU data obtained from extreme manoeuvring tests, the robustness of the drop-off mechanism during flight can be confirmed.

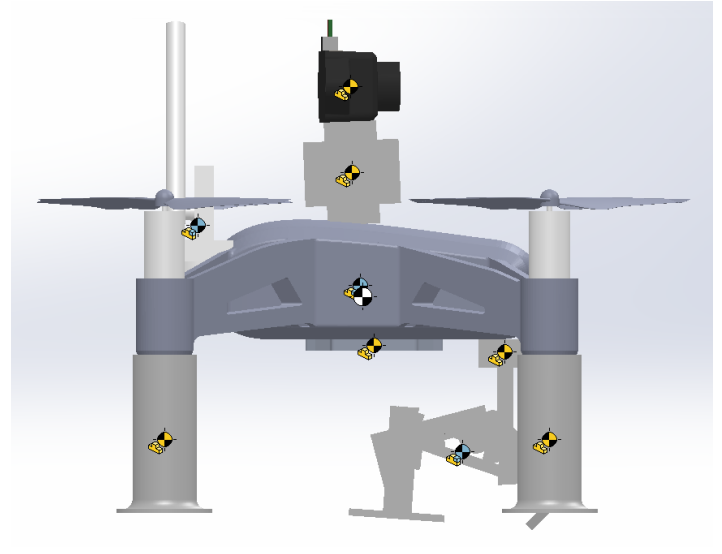
6.2 Further improvement of the design

The fact that the objective of reaching TRL level 4 has been met and success rate reached 70% for the integrated test has proved that the designed mechanisms are valid to the second stage of the mission profile, operating the four tasks. However, the project did not reach the stage of testing a run through of the entire mission profile. It is worth pointing out that the weight and CG restrictions were based on manual controllability; hence, there is a possibility for the restrictions to get tighter after testing the navigation performance on the integrated drone. The maximum flight time of DJI Tello is 13 minutes^[5] without attachments, and mounting sensors and mechanisms leads to a decrease in flight size. Shortened flight time reduces the range which the drone can perform its tasks. Therefore, further weight optimisation is desirable.

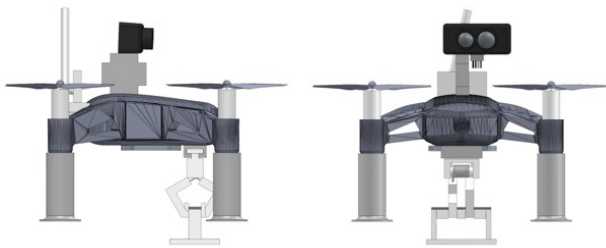
The leg extension part was something the Hardware team could have invested further energy into. The total mass of the four weights are 9.2 g, close to a quarter of the overall added weight. The leg extension were not expected to damp the landing motion since the damping may lead to decrease in the amount of energy that can be transferred into the cone penetration energy.



(a) Soil sampling mechanism using cones



(b) Soil sampling mechanism using claws



(c) Drop-off mechanism



(d) Leaf sampling mechanism

Figure 17: Side views and some forward views of assembled CAD of the Tello drone with all sensors and mechanisms integrated.

Table 5: Summary of how parts were integrated. Shift in centre of gravity after mounting mechanisms and sensors are listed at the bottom of the table. Note that for the drop off and the pickup mechanisms, the payload is included, since the CG shift is larger with payload. "Net moment" refers to clockwise moment viewing from starboard of the drone.

Component	Leaf	Soil	Drop-off	Pick-up	Connected to?	Connected via?
LIDAR	✓	✓	✓	✓	LIDAR stand	blu tack
LIDAR stand	✓	✓	✓	✓	drone's top surface	superglue
Arduino	✓	✓	✓	✓	drone's bottom surface	double-sided tape
FPV camera		✓	✓	✓	camera hanger	blu tack
Extended leg		✓	✓	✓	built-in leg	hollow
Actuator	✓				LIDAR stand	iron wire
Capasitive sensor	✓				drone's bottom surface	blu tack
Silicone rubber tube	✓				drone's top surface	blu tack
Horizontal bar	✓				propeller guards	superglue
leaf-sampling clip	✓				Horizontal bar	superglue
Universal socket		✓	✓	✓	drone's bottom surface	superglue
Soil-sampling clip		✓			Universal socket	sliding in
Drop-off mechanism			✓		Universal socket	sliding in
Pick-up mechanism				✓	Universal socket	sliding in
Total extra weight, g	32.3	42.0	43.7	44.7		
Net longitudinal cg shift, mm	6.85	0.46	0.31	0.75		
Net vertical cg shift, mm	4.73	2.42	2.35	0.42		
Net moment, cm-gf	22.1	1.9	1.4	3.4		

However, the lack of flexibility of the legs makes landing on rough or inclined surfaces challenging. Introducing and modifying the landing system with shock absorbers developed by Zhang et al.[18] may lead to both reduction of weight and add robustness of the landing procedure. The adjustment of the length of the leg extensions will still remain as a challenge, since the damping behaviour may invoke the need to make further adjustments on the mechanisms.

The payload can be improved in a similar manner, namely by changing the shape of the bottom surface or add extensions to stabilise its pose after drop-off. Recalling that the main cause of failure of the integrated drop-off test was due to the payload flipping over after release, the success rate of the mission may increase. Additionally, it may become possible for the drop-off mission to be performed on various surface conditions.

7 Conclusion

The main objective of the Hardware team was to design modular mechanisms that can be integrated onto DJI Tello, and operate tasks related to environmental monitoring, which is becoming increasingly demanding over the past years. Through literature reviews, brainstorming, mathematical modelling and experimental testing of prototypes, four mechanisms, a modular attachment base and other supporting parts

were designed. The designed mechanisms reached TRL level 4 on multiple tasks included in the mission profile: namely soil sampling collection, drop-off and pick-up of the sensor package. Additionally, when mounted on the drone, the mechanisms achieved a success rate of 70% or greater. However, the project did not reach the stage of testing the entire mission profile. Further improvements of the hardware are in sight, and is left for future investigations. In conclusion, the work of the Hardware team has been a success in a sense that the objectives were met.

References

- [1] S. Manfreda, M. McCabe, P. Miller, R. Lucas, V. P. Madrigal, G. Mallinis, E. B. Dor, D. Helman, L. Estes, G. Ciraolo, J. Müllerová, F. Tauro, M. de Lima, J. de Lima, A. Maltese, F. Frances, K. Caylor, M. Kohv, M. Perks, G. Ruiz-Pérez, Z. Su, G. Vico, and B. Toth, “On the use of unmanned aerial systems for environmental monitoring,” *Remote sensing (Basel, Switzerland)*, vol. 10, p. 641, Apr 20, 2018.
- [2] I. Leifer, W. J. Lehr, D. Simecek-Beatty, E. Bradley, R. Clark, P. Dennison, Y. Hu, S. Matheson, C. E. Jones, B. Holt, M. Reif, D. A. Roberts, J. Svejksky, G. Swayze, and J. Wozencraft, “State of the art satellite and airborne marine oil spill remote sensing: Application to the bp deepwater horizon oil spill,” *Remote Sensing of Environment*, vol. 124, pp. 185–209, 2012.
- [3] H. T. Lally, I. Connor, O. P. Jensen, and C. T. Graham, “Can drones be used to conduct water sampling in aquatic environments? a review,” *The Science of the total environment*, vol. 670, pp. 569–575, Jun 20, 2019.
- [4] A. Farinha, R. Zufferey, P. Zheng, S. F. Armanini, and M. Kovac, “Unmanned aerial sensor placement for cluttered environments,” *IEEE robotics and automation letters*, vol. 5, pp. 6623–6630, Oct 2020.
- [5] RYZE, “Tello specs.” Available from: <https://www.ryzerobotics.com/tello/specs>. [Accessed 14th June 2021].
- [6] G. John and S. Messenger, “Ash dieback: Deadly tree fungus spreading ‘more quickly’.” Available from: <https://www.bbc.com/news/uk-wales-47483197>, 2019. [Accessed 14th June 2021].
- [7] F. Research, “Soil contamination - issues and practical considerations.” Available from: <https://www.forestresearch.gov.uk/tools-and-resources/fthr/urban-regeneration-and-greenspace-partnership/greenspace-in-practice/practical-considerations-and-challenges-to-greenspace/soil-contamination-issues-and-practical-c> 2019. [Accessed 14th June 2021].
- [8] NASA, “Technology readiness level.” Available from: https://www.nasa.gov/directorates/heo/scan/engineering/technology/technology_readiness_level, Oct 28, 2012. [Accessed 14th June 2021].
- [9] SoilsWorldwide, “Soil classification system of england and wales.” Available from: http://www.soilsworldwide.net/index.php/Soil_classification_system_of_England_and_Wales. [Accessed 14th June 2021].
- [10] M. Kovac, “The bioinspiration design paradigm: A perspective for soft robotics,” *Soft robotics*, vol. 1, pp. 28–37, 2014.
- [11] I. Patringenaru, “From the deep sea to deep space: Sea urchin’s teeth inspire new design for space exploration device.” Available from: https://ucsdnews.ucsd.edu/pressrelease/from_the_deep_sea_to_deep_space_sea_urchins_teeth_inspire_new_design_for_sp, May 3, 2016. [Accessed 24th June 2021].
- [12] M. Shahinpoor, “Biomimetic robotic venus flytrap (dionaea muscipula ellis) made with ionic polymer metal composites,” *Bioinspiration & biomimetics*, vol. 6, p. 046004, Dec 01, 2011.
- [13] G.-J. Xiong, J.-J. Chen, J.-H. Wang, and M.-G. Li, “New axisymmetric slip-line theory for metal and its application in indentation problem,” *Journal of Engineering Mechanics*, vol. 145, -12 2019.
- [14] TheEngineeringToolBox, “Density of materials.” Available from: https://www.engineeringtoolbox.com/density-materials-d_1652.html. [Accessed 14th June 2021].
- [15] S. Ruiz, D. Or, and S. J. Schymanski, “Soil penetration by earthworms and plant roots—mechanical energetics of bioturbation of compacted soils,” *PloS one*, vol. 10, no. 6, p. e0128914, 2015.

- [16] R. F. Bishop, R. Hill, and N. F. Mott, “The theory of indentation and hardness tests,” *Proceedings of the Physical Society*, vol. 57, pp. 147–159, may 1945.
- [17] J. K. A. Langowski, P. Sharma, and A. L. Shoushtari, “In the soft grip of nature,” *Science robotics*, vol. 5, no. 49, 2020.
- [18] K. Zhang, P. Chermprayong, D. Tzoumanikas, W. Li, M. Grimm, M. Smentoch, S. Leutenegger, and M. Kovac, “Bioinspired design of a landing system with soft shock absorbers for autonomous aerial robots,” *Journal of field robotics*, vol. 36, pp. 230–251, Jan 2019.

A Supplementary data

A.1 Magnet precision test

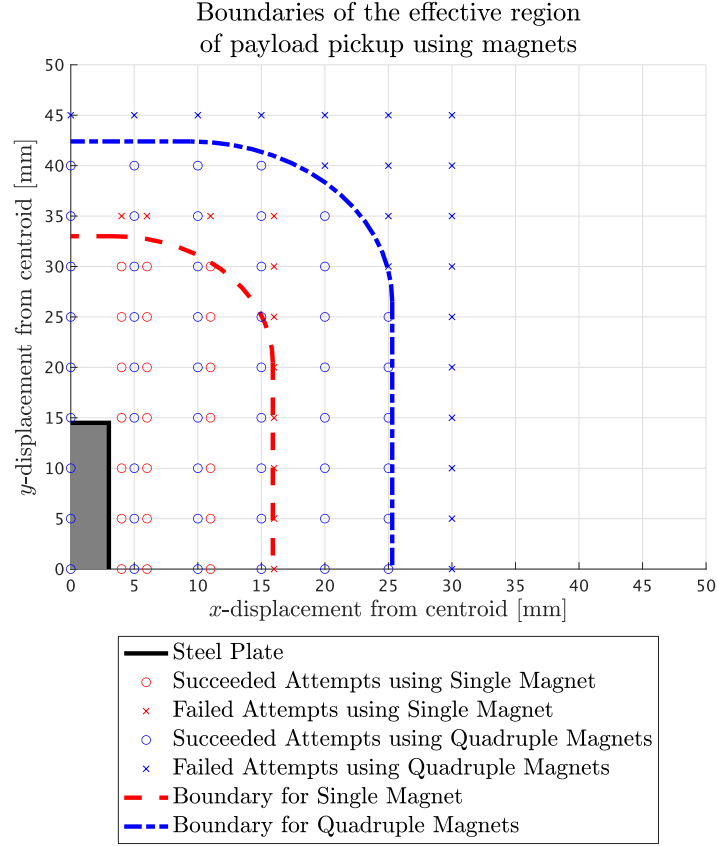


Figure A.1: The boundaries of the effective region and the individual attempts of the pick up mechanism. The boundaries shown on the figure represents the range of the centre of the pick up mechanisms, in local x, y coordinates, that succeeded in picking up the payload.

B Details of the theory used for soil sampling design

The approach of calculating indentation energy by Ruiz was rather simple, and substituting soil parameters listed in Table 1 into the following equations:

$$\begin{aligned}
 P_L &= s_u \left(1 + \ln \left(\frac{G}{s_u} \right) \right) \\
 \sigma_r &= P_L - 2s_u \ln \left(\frac{r}{r_b} \right) \\
 D &= \frac{\sin \left(\frac{\pi - \alpha}{2} \right) + m \sin(\pi - \alpha)}{\cos \left(\frac{\pi - \alpha}{2} \right) - \cos(\pi - \alpha)} \\
 \sigma_z &= \sigma_r + \\
 &\quad s_u \left(\pi + 2\alpha + \sin^{-1}(m) + \frac{D}{2} + m \cot \alpha - \sqrt{1 - m^2} - 1 \right) \\
 F_z &= 2\pi \int_0^{\delta_0 \tan \alpha} r \sigma_z dr \\
 U_{pen} &= \int F_z dz
 \end{aligned}$$

It is worth noting that wide range of value of s_u was given for minimal clay percentage in Ruiz et al.[15], a value closed to the geometrical mean of the limits were taken in Table 1.

C Matlab Code

C.1 Function used to analyse landing data from IMU

```

1
2 function [a, v, h, KE, t, KEI] = LandingDataProcess(time, acc, Offset, End, needKE)
3
4 % Constants
5 g = 9.806; % m/s^2, gravitational acceleration (roughly at latitude of 45)
6 m = 105e-3; % kg, mass of the drone and its attachments
7
8
9
10 Nstart = find(abs(acc - Offset) > 0.04, 1) - 3;
11 if End == 0
12     Nend = find(acc > -0.16, 2);
13 else
14     Nend = End;
15 end
16
17 % Time
18 t = time(Nstart:Nend(end)) - time(Nstart);
19
20 % Acceleration
21 % Subtract offset and then convert into m/s^2
22 a = (Offset - acc(Nstart:Nend(end))) * g;
23 % Smoothen
24 % a = smooth(a, 'loess');
25
26 % Velocity
27 v = cumtrapz(t, a);
28
29 % Height
30 h = cumtrapz(t, v);
31
32 % Kinetic energy
33 KE = m/2 * v.^2;
34
35 KEI = KE( find( abs(h(300:end)) == min(abs(h(300:end))) ) );
36
37 %% Plot
38
39 switch needKE
40     case false
41         Nplot = 3;
42     case true
43         Nplot = 4;
44 end

```

```

45
46 fig = figure;
47
48 subplot(Nplot,1,1);
49 plot(t,a)
50 grid on
51 title('Acceleration','interpreter','latex')
52 ylabel('Acceleration, [m/s$^2$]','FontSize',12,'Interpreter','latex')
53
54 subplot(Nplot,1,2);
55 plot(t,v)
56 grid on
57 title('Velocity','interpreter','latex')
58 ylabel('Velocity, [m/s]','FontSize',12,'Interpreter','latex')
59
60 subplot(Nplot,1,3);
61 plot(t,h)
62 grid on
63 title('Height','interpreter','latex')
64 ylabel('Height, [m]','FontSize',12,'Interpreter','latex')
65
66 if Nplot == 4
67     subplot(4,1,4);
68     plot(t,KE)
69     grid on
70     title('Kinetic energy','interpreter','latex')
71     ylabel('Kinetic energy, [J]','FontSize',12,'Interpreter','latex')
72 end
73
74 xlabel('Time, [s]','FontSize',16,'Interpreter','latex')
75 end

```

D Candidates of mechanisms and Pugh diagrams

Figure D.2: Pugh table for choosing gripping mechanisms. Magnetic gripper achieved the highest score.

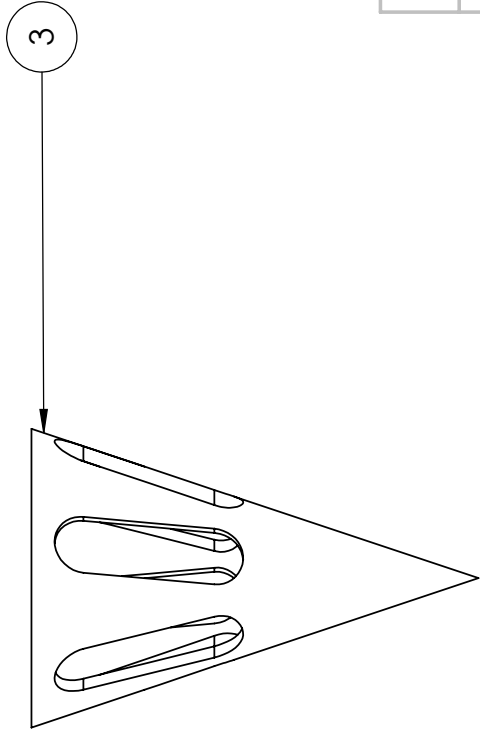
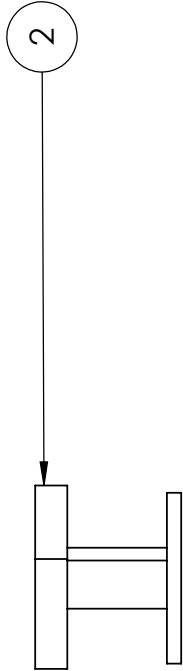
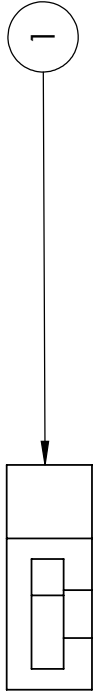
	Desired payload traits	Cost effectiveness	Weight effectiveness	Robustness	Ease of implementation (including manufacturability)	Ease of pick-up	Creativity / ingenuity	Energy effectiveness	Payload restrictions	Scalability	Score
(Weightage)		1	2	1	1	1	0.3	1	2	1	
Avian-inspired claw		4	5	4	2	4	4	2	4	5	3.902913
Gecko inspired soft passive gripper	Flat & clean surface	3	5	1	3	2	4	4	1	5	3.029126
Gecko-inspired mechanism using adhesive & pneumatic actuator	Preferably clean surface	5	1	2	2	4	4	3	4	3	2.932039
Origami vacume gripper	NA for thin/large objects	4	3	2	3	3	4	2	3	5	3.126214
Magnetic gripper	Preferably clean surface/ ferous materials	5	5	4	5	2	1	4	3	5	4.009709
Granular jamming	Not suitable for flat wide/thin objects	5	1	2	4	2	2	2	3	3	2.582524
Self-sealing suction cup	Flat clean surface	4	3	3	2	3	3	2	3	3	2.902913
Retractable tether	Requires additional packaging/accessory by means of attaching the payload to one end of the tether	5	3	1	4	2	1	2	1	5	2.650485
Soft hybrid suction cup	Preferably clean and flat	4	3	4	3	3	3	1	5	3	3.38835
Pneumatically powered elastic gripper	Can pick up irregular shaped objects	4	2	4	3	4	4	3	3	2	3.029126

E CAD Drawings

6 5 4 3 2 1

D

D



Num	Part Name	Qty
1	Female Trail	1
2	Male Trail for Cone	1
3	Cone	1

C

C

B

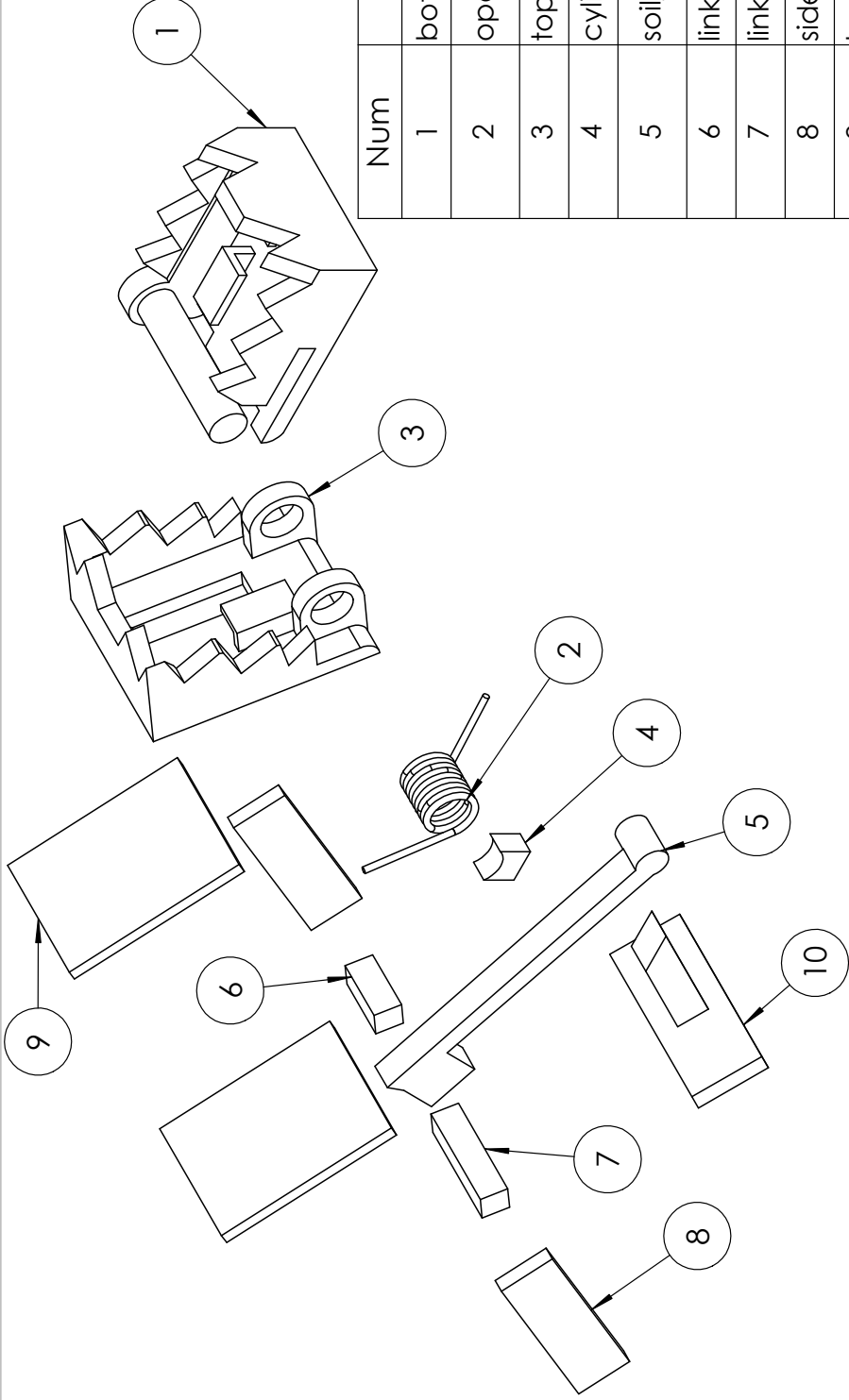
B

A

A

All tolerances are +/- 0.1 unless stated otherwise			A4	
Imperial College London	Scale:	2:1	Title: Soil Sampling Assembly (Cone)	
	Units:	mm	Drawn by: ADD3	
Material: PLA	Date: 11/06/2021		1 of 1	

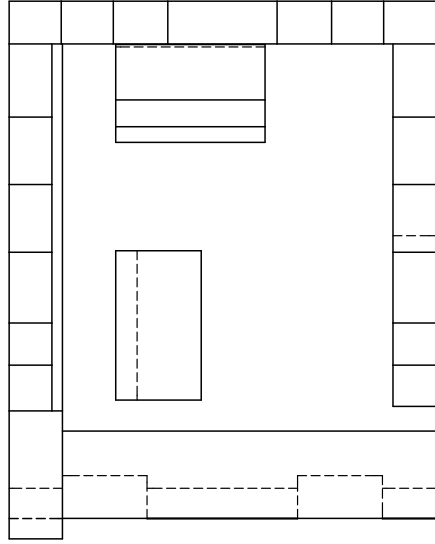
6 5 4 3 2 1



Num	Part Name	Qty
1	bottomhalf	1
2	opened torsional spring	1
3	tophalf	1
4	cylinder_support	1
5	soil_supportive_internal	1
6	linkage7	1
7	linkage12	1
8	side_support	2
9	bottom_pad	2
10	drone_connection	1

All tolerances are +/- 0.1 unless stated otherwise					A4
Imperial College London	Scale:	1.5:1			Soil Claw
	Units:	mm			
	Date:	12/06/2021			
Material: PLA	Drawn by:			ADD3	1 of 1

6 5 4 3 2 1

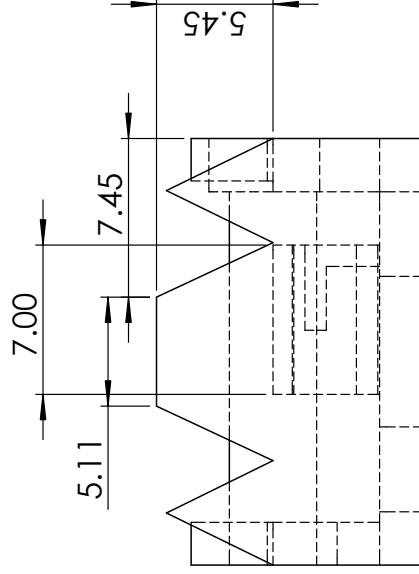
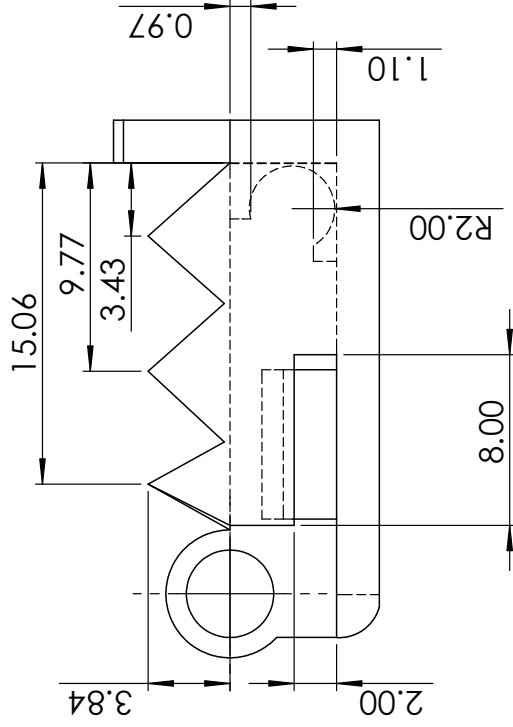


D

C

B

A



All tolerances are +/- 0.1 unless stated otherwise

A4

Imperial College
London

Scale:
2:1

Title:

Bottom half

Units:
mm

Material:

PLA

Date:

12/06/2021

Drawn by:

ADD3

1 of 1

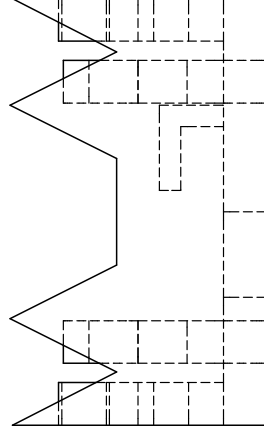
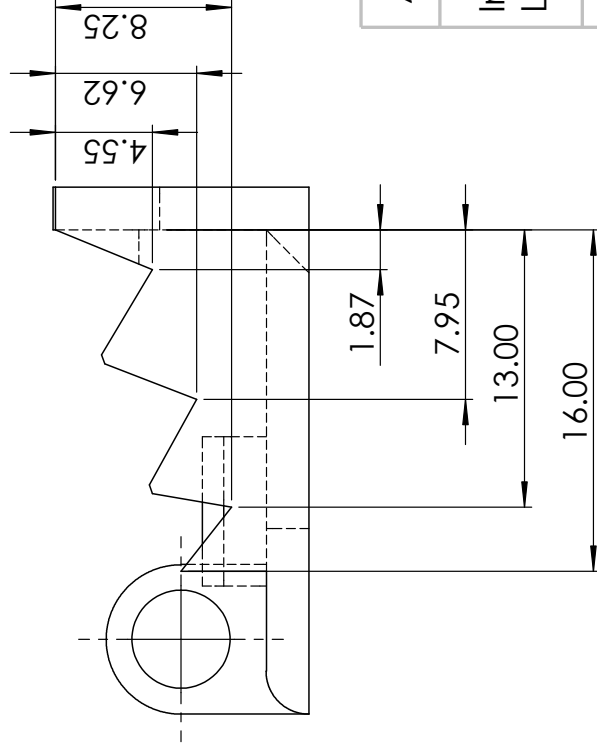
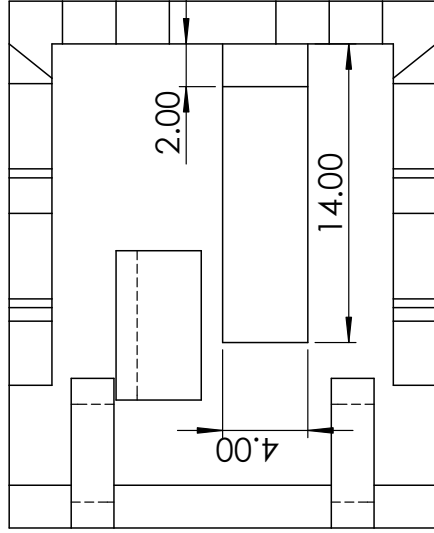
6 5 4 3 2 1

D

C

B

A



All tolerances are +/- 0.1 unless stated otherwise

A4

Imperial College
London

Scale:
3:1

Title:

Tophalf

Material:

PLA

Units:

mm

Date:

12/06/2021

Drawn by:

ADD3

1 of 1

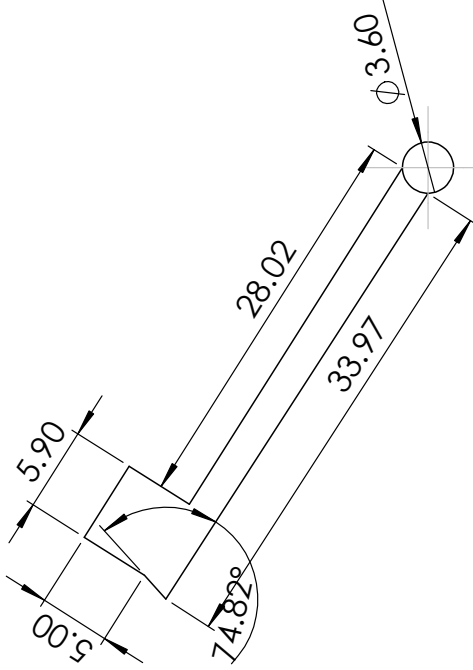
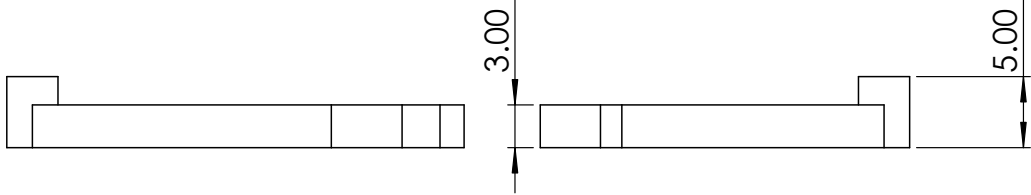
6 5 4 3 2 1

D

C

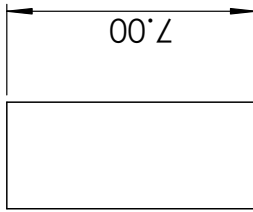
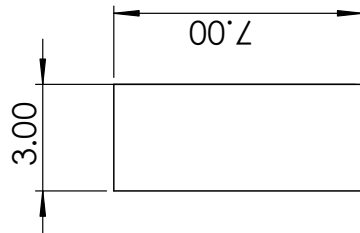
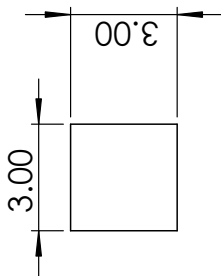
B

A



All tolerances are +/- 0.1 unless stated otherwise			A4	
Imperial College London	Scale:	2:1	Title: soil support internal	
	Units:	mm	Drawn by: ADD3	
Material: PLA		Date:	12/06/2021	
			1 of 1	

6 5 4 3 2 1



D C B A

All tolerances are +/- 0.1 unless stated otherwise			A4	
Imperial College London	Scale:	Linkage7		
	5:1			
	Material: PLA	Units:	Drawn by: ADD3	
mm				
		Date:	1 of 1	
	12/06/2021			

6 5 4 3 2 1

D

D

C

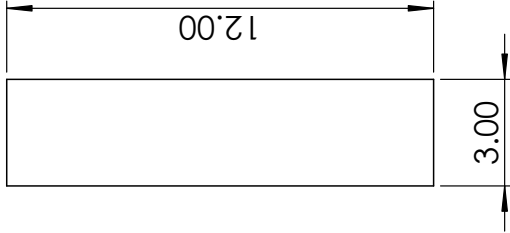
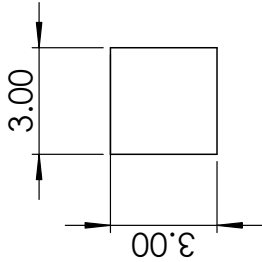
C

B

B

A

A



All tolerances are +/- 0.1 unless stated otherwise			A4	
Imperial College London	Scale: 5:1		Title: Linkage12	
	Units: mm			
	Material: PLA	Date: 12/06/2021		Drawn by: ADD3
		1 of 1		

6 5 4 3 2 1

D

D

C

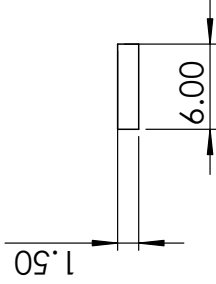
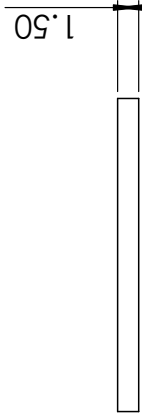
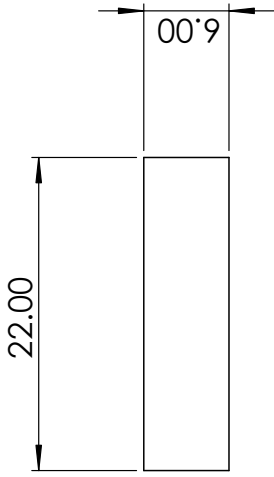
C

B

B

A

A



All tolerances are +/- 0.1 unless stated otherwise			A4	
Imperial College London	Scale: 2:1	Title: Side support		1 of 1
	Units: mm	Drawn by: ADD3		
	Material: PLA		Date: 12/06/2021	

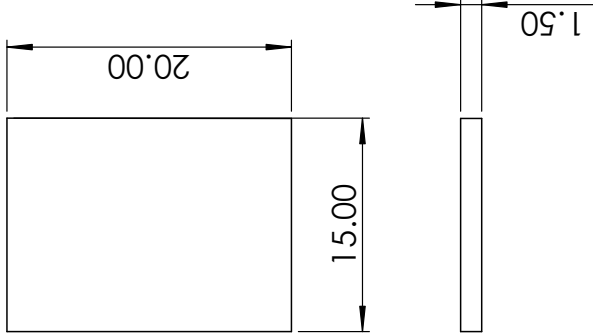
6 5 4 3 2 1

D

C

B

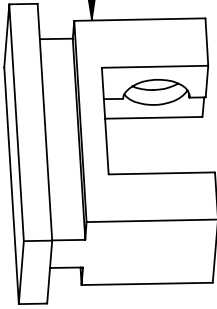
A



All tolerances are +/- 0.1 unless stated otherwise			A4	
Imperial College London	Scale: 2:1		Title: Bottom pad	
	Units: mm			
	Material: PLA	Date: 12/06/2021		Drawn by: ADD3
		1 of 1		

6 5 4 3 2 1

D



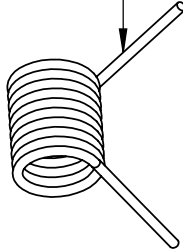
4

D



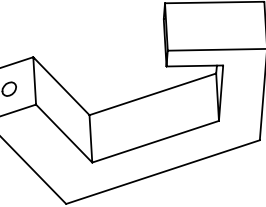
3

C



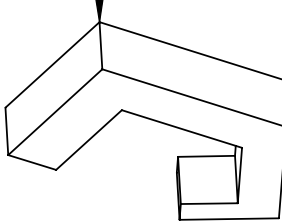
2

C



1

B



B

Num	Part Name	Qty
1	Claw	2
2	Torsional Spring	1
3	Spring Support	1
4	Male Trail for Drop Off	1

All tolerances are +/- 0.1 unless stated otherwise

A4

Imperial College
London

Scale:

2:1

Title:

Drop-Off Mechanism

Material:

PLA & Stainless
Steel

Units:

mm

Date:

10/06/2021

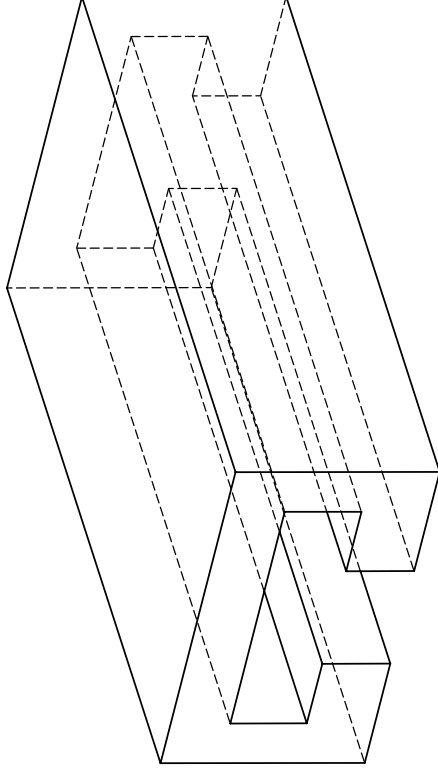
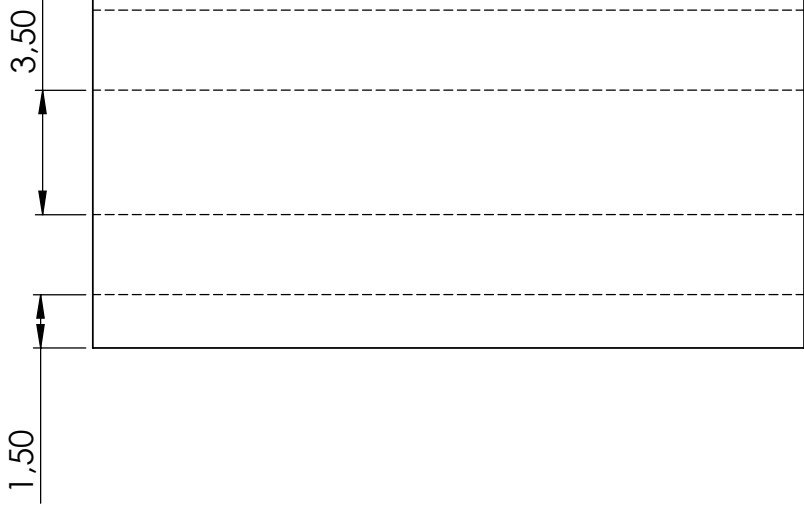
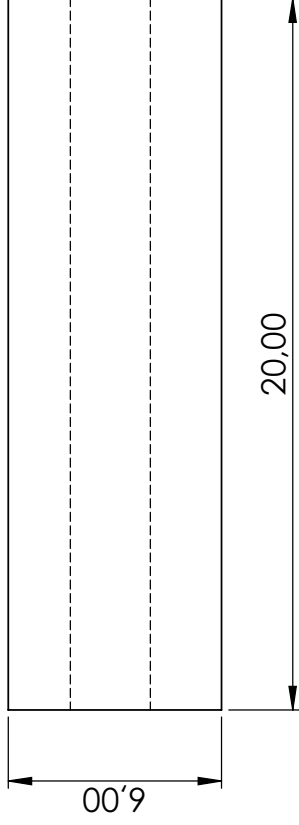
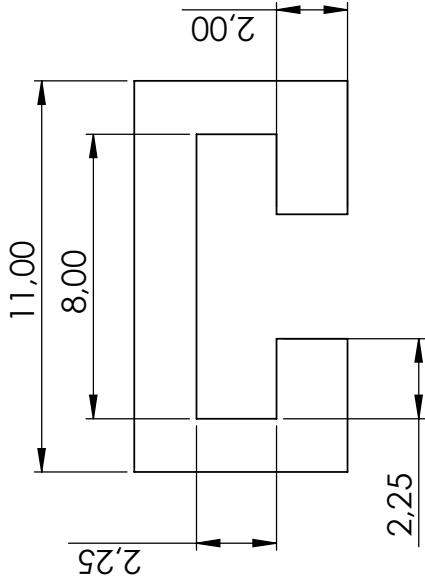
Drawn by:

ADD3

1 of 1

A

6 5 4 3 2 1



All tolerances are +/- 0.1 unless stated otherwise			A4	
Imperial College London	Scale:	5:1	Title: Female Trail	
	Units:	mm		
Material: PLA		Date:	Drawn by:	
		10/06/2021	ADD3	
			1 of 1	

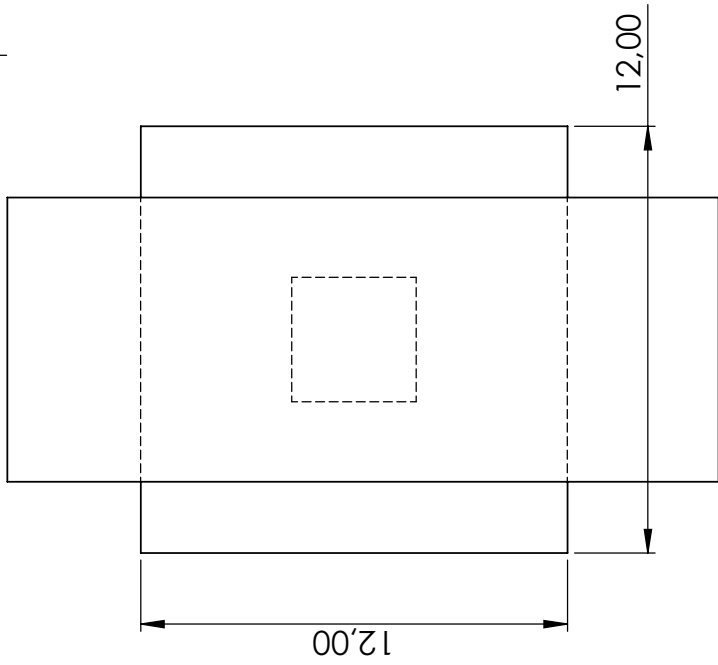
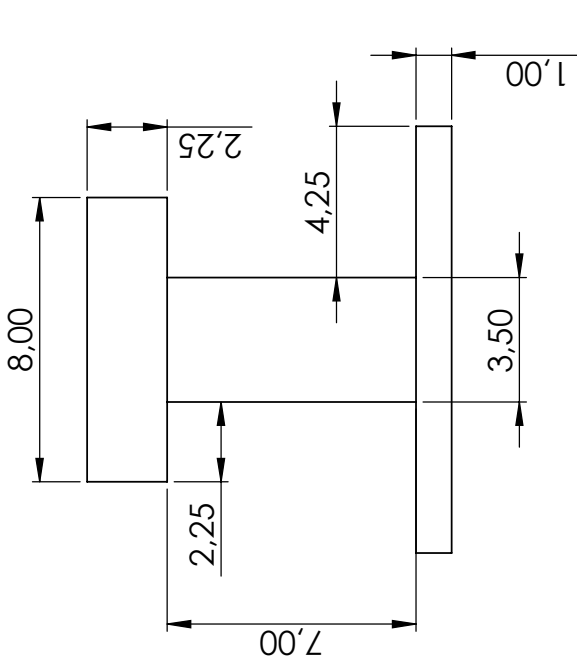
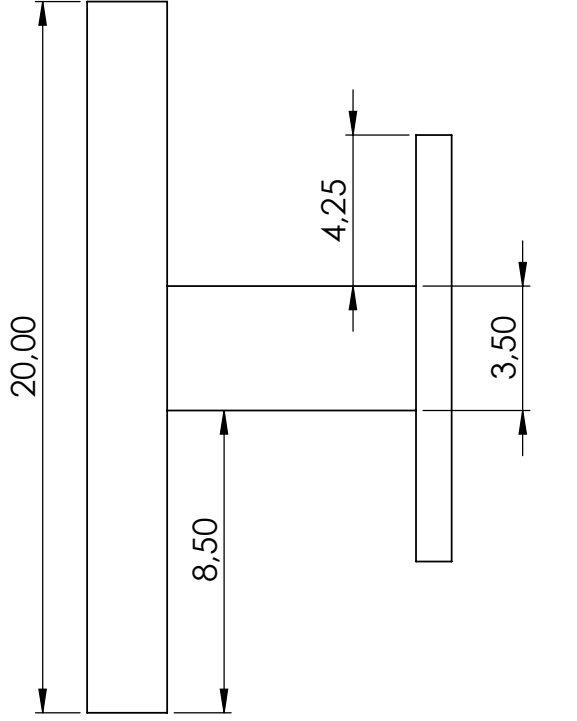
6 5 4 3 2 1

D

C

B

A



All tolerances are +/- 0.1 unless stated otherwise

A4

Imperial College
London

Scale:

5:1

Units:

mm

Title:

Male Trail for Pick Up

Material:

PLA

Date:

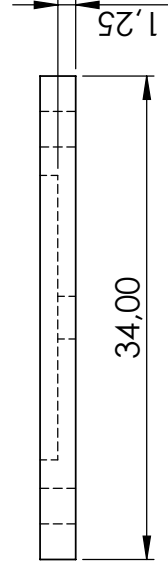
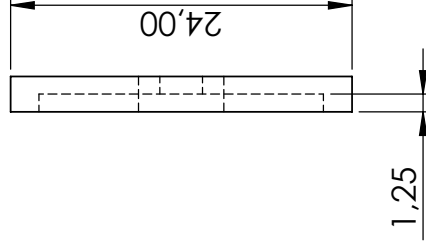
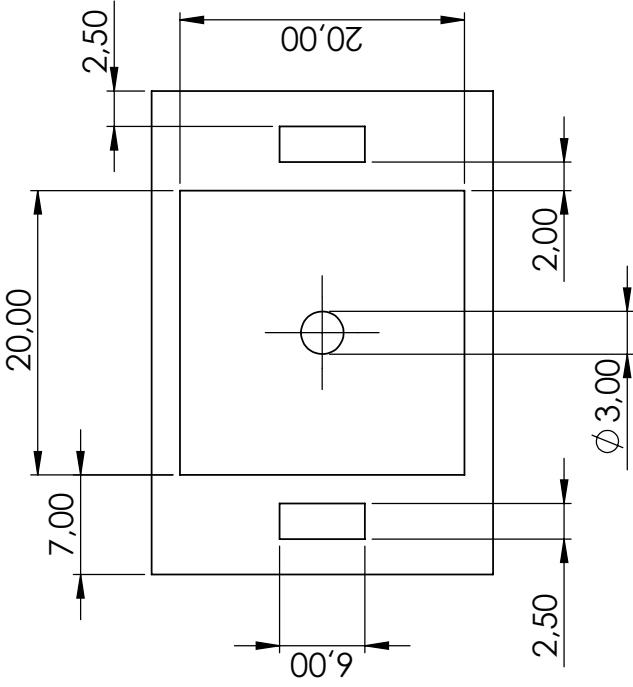
10/06/2021

Drawn by:

ADD3

1 of 1

6 5 4 3 2 1



All tolerances are +/- 0.1 unless stated otherwise			A4	
Imperial College London	Scale:	Title: Payload Base		
	2:1			
	Units:	mm		
Material: PLA	Date:	Drawn by: ADD3		1 of 1
	10/06/2021			

6 5 4 3 2 1

D

C

B

A

D

C

B

A

All tolerances are +/- 0.1 unless stated otherwise			A4	
Imperial College London	Scale: 2:1		Title: Steel Plate	
	Units: mm			
	Material: Steel	Date: 10/06/2021		Drawn by: ADD3

3 2 1

4

5

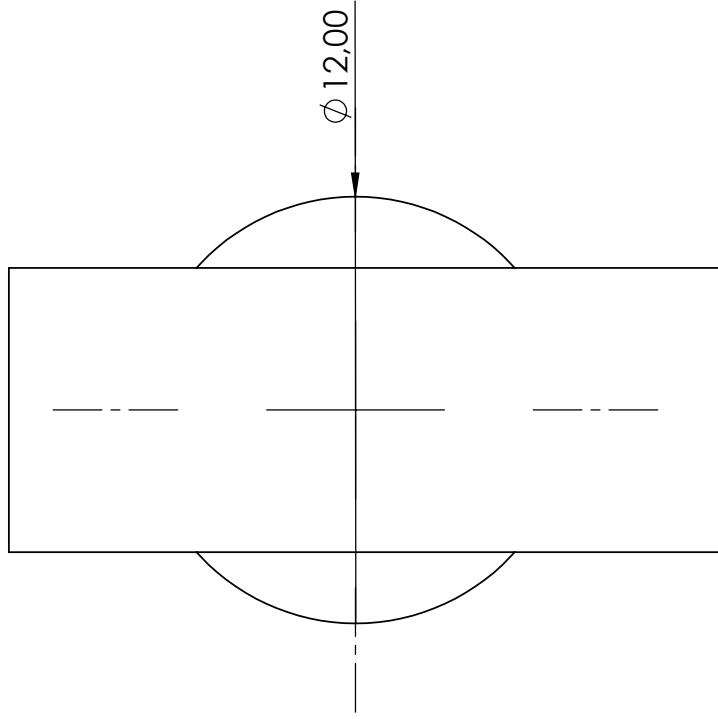
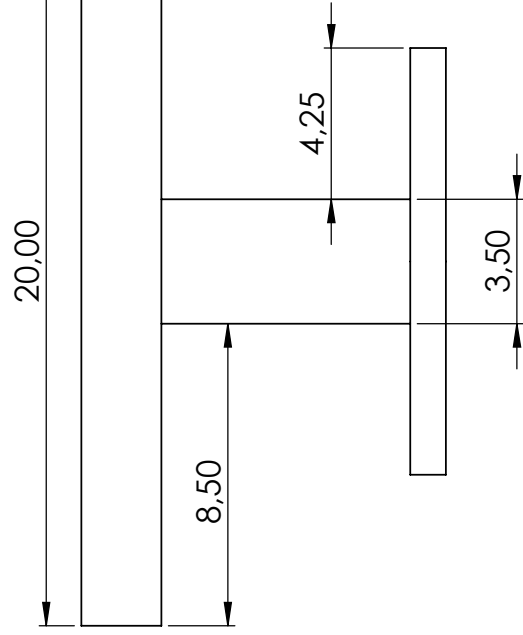
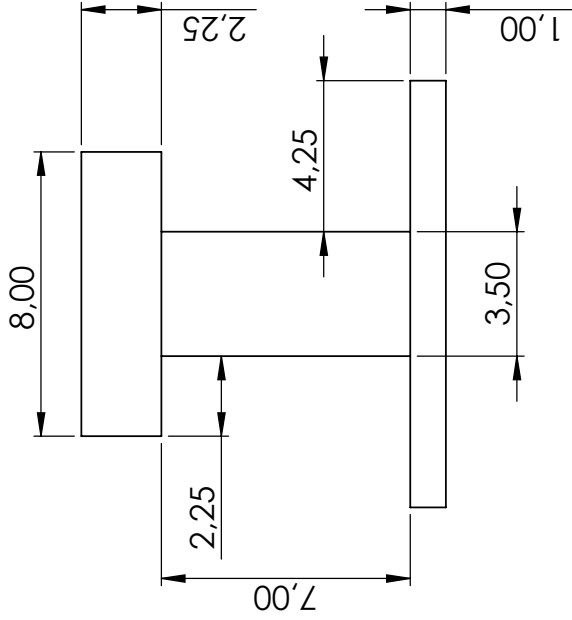
6 5 4 3 2 1

D

C

B

A



All tolerances are +/- 0.1 unless stated otherwise

A4

Imperial College
London

Scale:

5:1

Units:

mm

Date:

10/06/2021

Title:

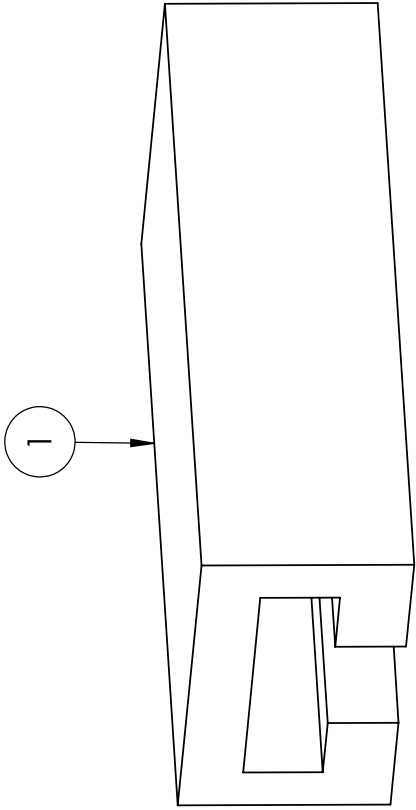
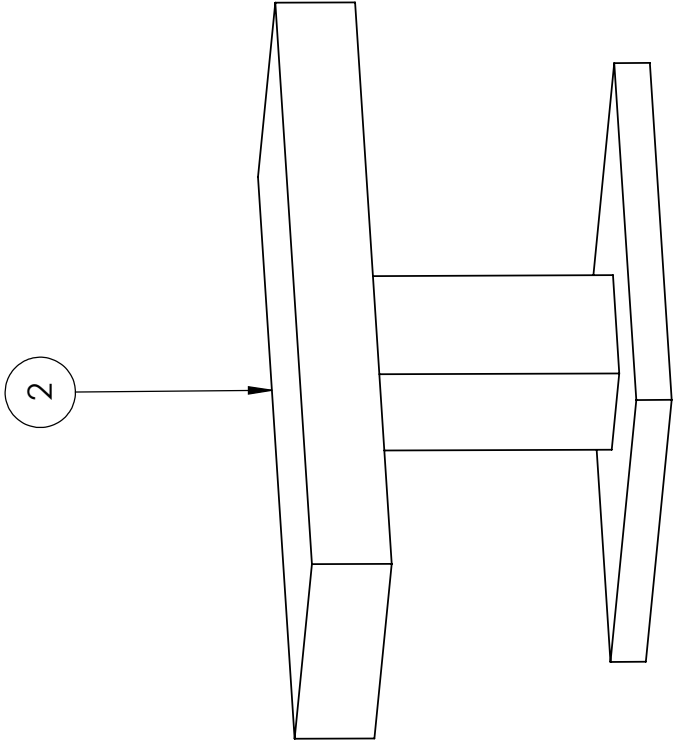
Male Trail for Cone

Drawn by:

ADD3

1 of 1

6 5 4 3 2 1



Num	Part Name	Qty
1	Male Trail for Pick Up	1
2	Female Trail	1

B

B

All tolerances are +/- 0.1 unless stated otherwise			A4
Imperial College London	Scale:	Title:	
	5:1	Pick Up Trail Assembly	
	Units:	Drawn by:	
Material: PLA	mm	ADD3	1 of 1
	Date:	10/06/2021	

A

A

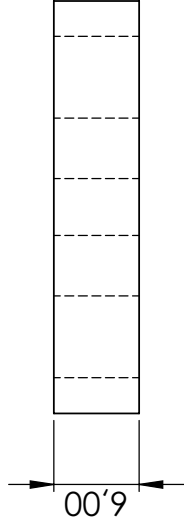
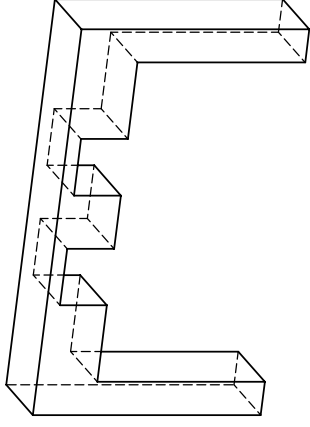
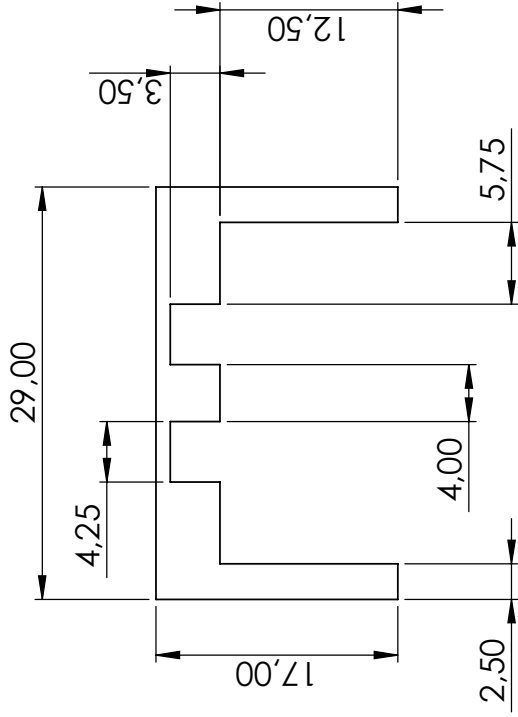
6 5 4 3 2 1

D

C

B

A



All tolerances are +/- 0.1 unless stated otherwise

A4

Imperial College
London

Scale:

2:1

Title:

Payload U Bar

Material:

PLA

Units:

mm

Date:

10/06/2021

Drawn by:

ADD3

1 of 1